NOVA

CODING CLUB

Innovate. Automate. Empower

Let’s build the future, one line of code at a time!

NOVA CODING CLUB

What is Coding?

Coding is giving instructions to a computer to make it do things.

Let’s build the future, one line of code at a time!

Why Learn to Code?

- Builds problem-solving & critical thinking
- Superpower for building things (websites, apps, games)
- Great career opportunities (software engineering, data science, AI)
- Fun & creative (you can code art, music, animations!)

How Does Coding Work?

Code → Instructions → Computer does something

Some programming languages:

- HTML (structure), CSS (style), JavaScript (interaction)
- Python (easy and popular for beginners)

How Coding Works (HTML + JavaScript)

```html
<!DOCTYPE html>
<html>
  <head>
    <title>My First Code</title>
  </head>
  <body>
    <h1>Hello, Coding Club!</h1>
    <p>Click the button to see some magic:</p>

    <button onclick="showMessage()">Click Me</button>

    <script>
      function showMessage() {
        alert("🎉 You just ran your first code!");
      }
    </script>
  </body>
</html>
```

Our Decree

This is how we roll in the Coding Club.

📌 **We start from scratch**
No experience? No problem. We all begin at step one.

📌 **We practice a lot**
The more we code, the better we get. Practice makes progress.

📌 **We work together**
We help each other out. We grow as a team.

📌 **We have fun**
Coding is serious magic—but it's also seriously fun.

📌 **Our goal is to learn**
We focus on growth, not perfection.

📌 **We gain a new skill**
By the end, you’ll have something cool to show off.

📌 **There are no silly questions**
Every question helps us learn. Ask away!

OUR ULTIMATE GOAL!

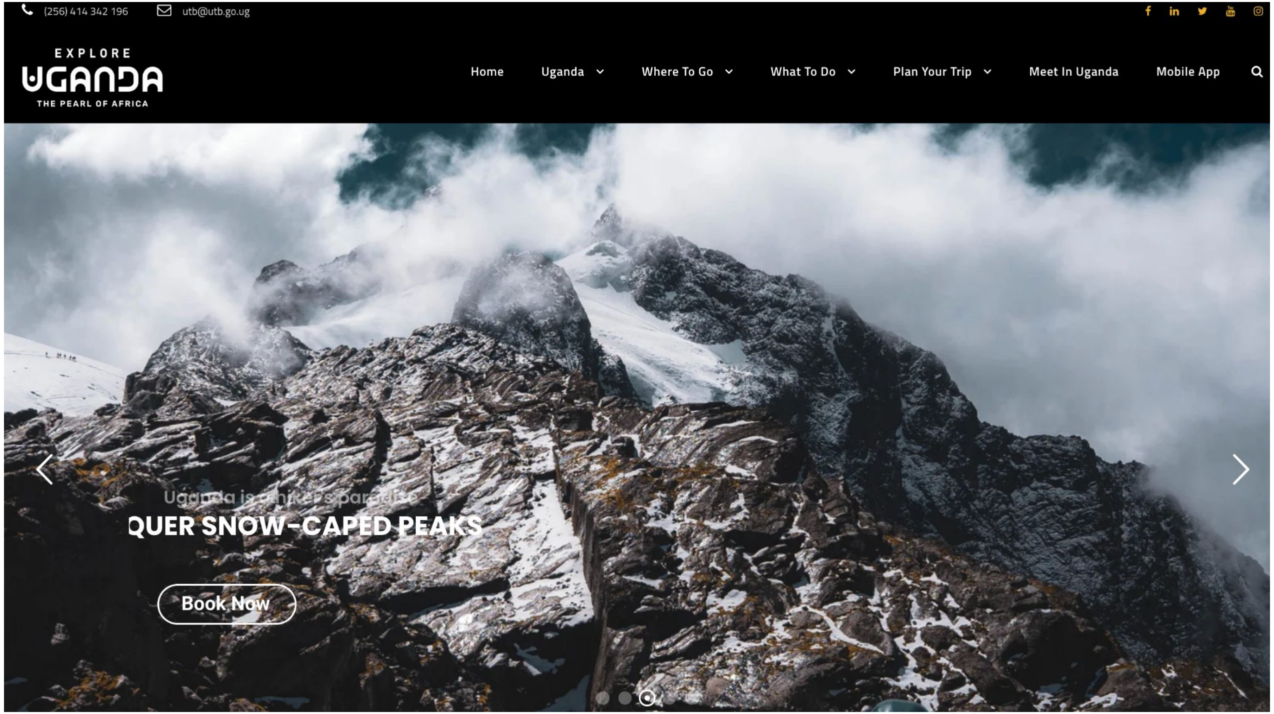
Build a quality static **website**

Gain a basic **professional skill**

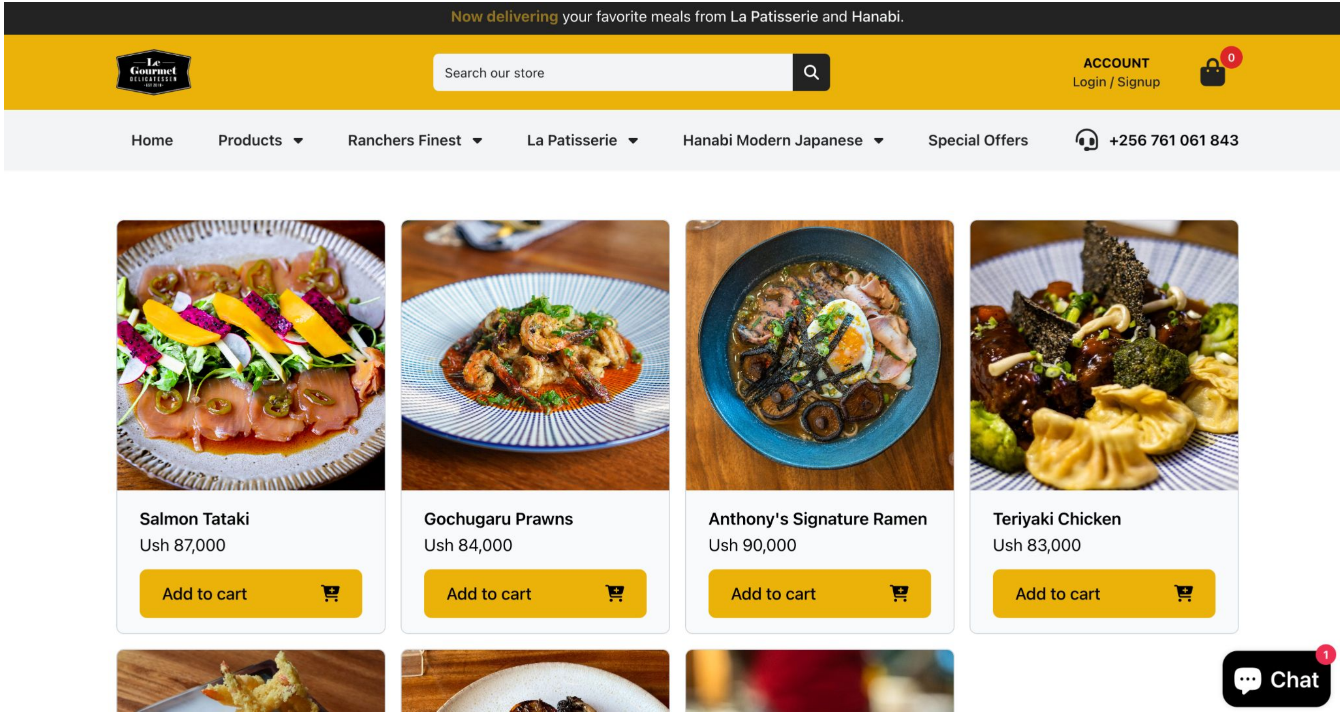
Marry up **technical** expertise with your **creative ability**

Improve our technical **confidence**

BASIC WEBSITE EXAMPLES



BASIC WEBSITE EXAMPLES



COURSE JOURNEY

MODULE 01	MODULE 02	MODULE 03	MODULE 04	MODULE 05	MODULE 06	MODULE 07	MODULE 08
HTML	CSS	Mini project	Intermediate HTML and CSS	JavaScript	Advanced Javascript	Github Pages Project work	Project presentations

01

HTML BASICS

What is HTML and what it is used for?

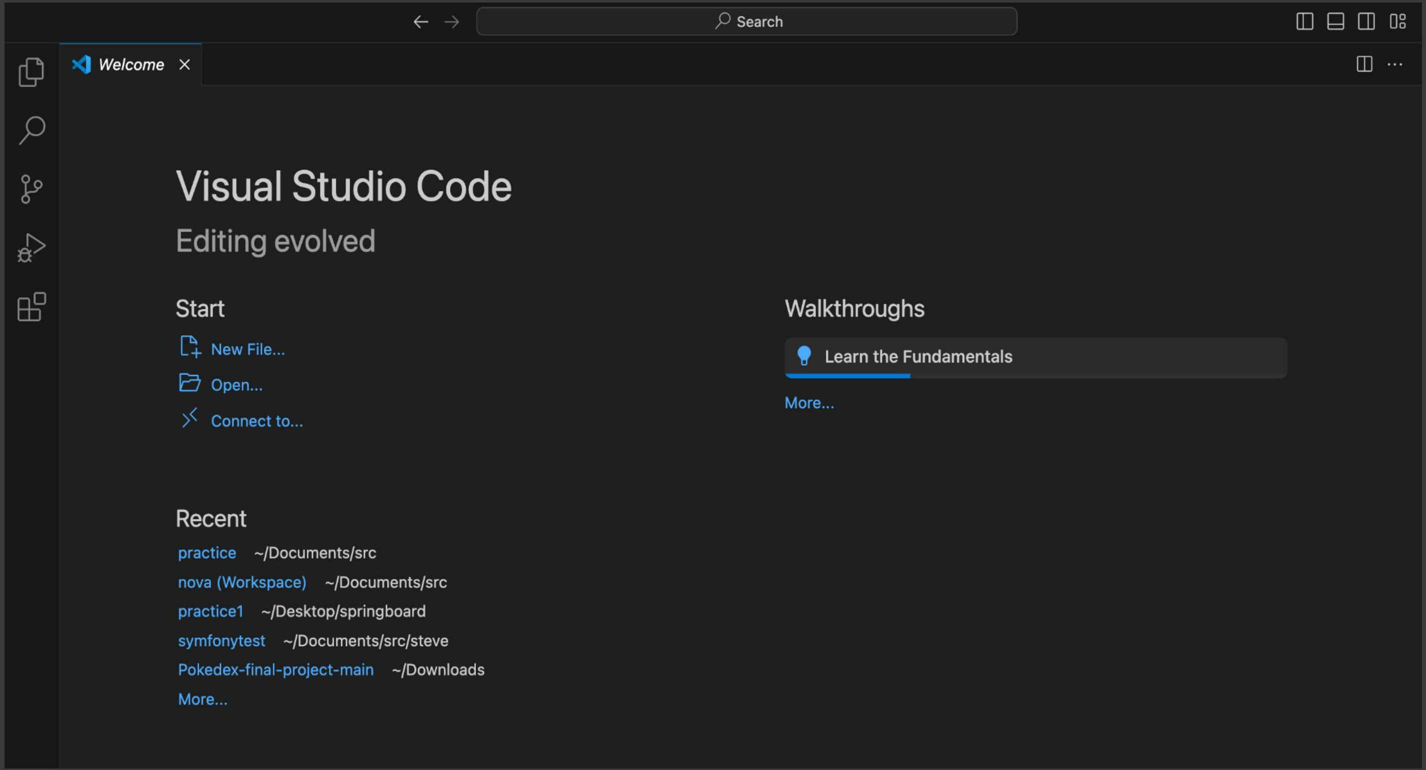
Learn basic HTML syntax

Learn how to structure and build a HTML page

Complete interesting practical exercises

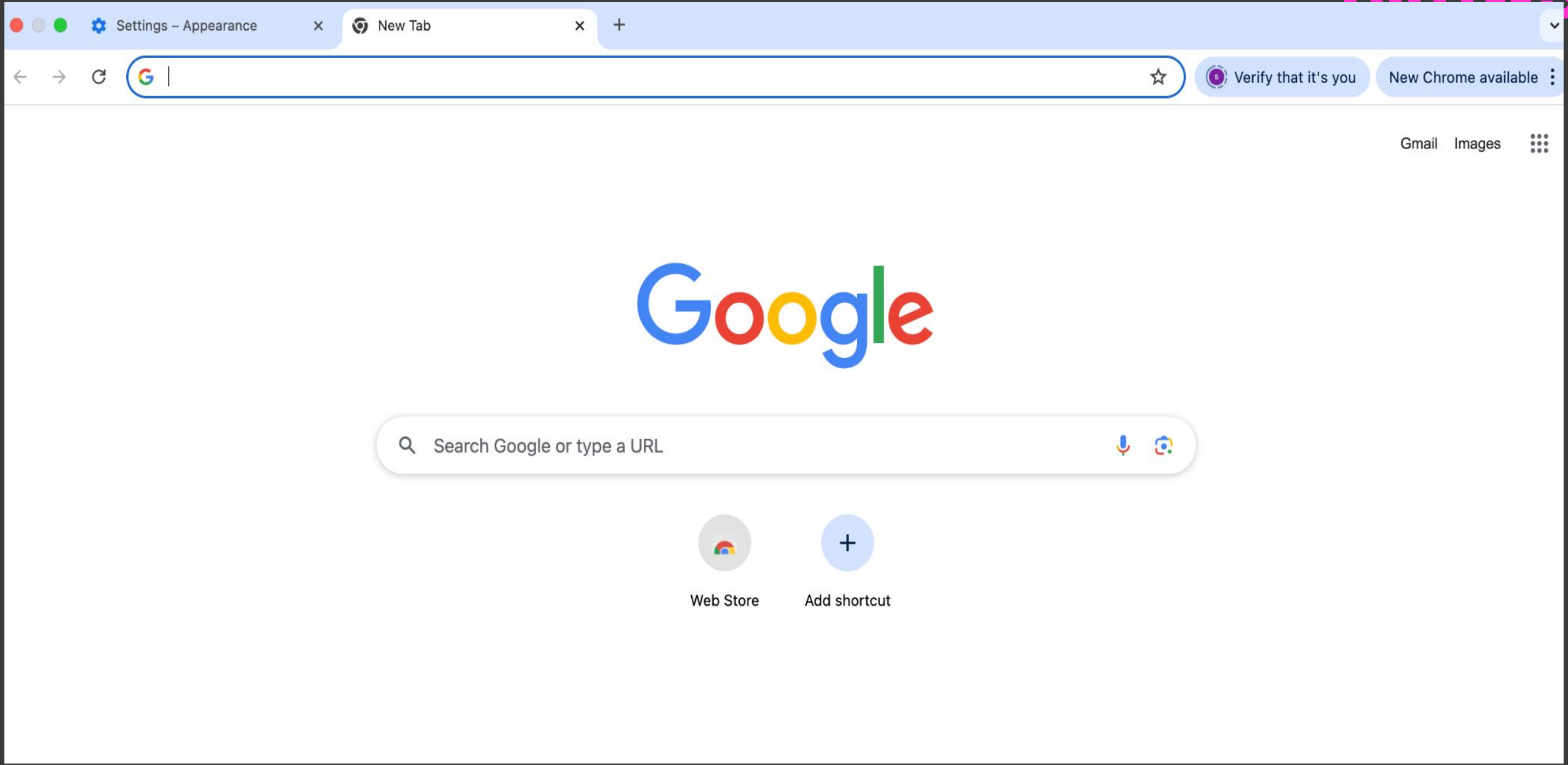
GETTING STARTED - TOOLS

VISUAL STUDIO



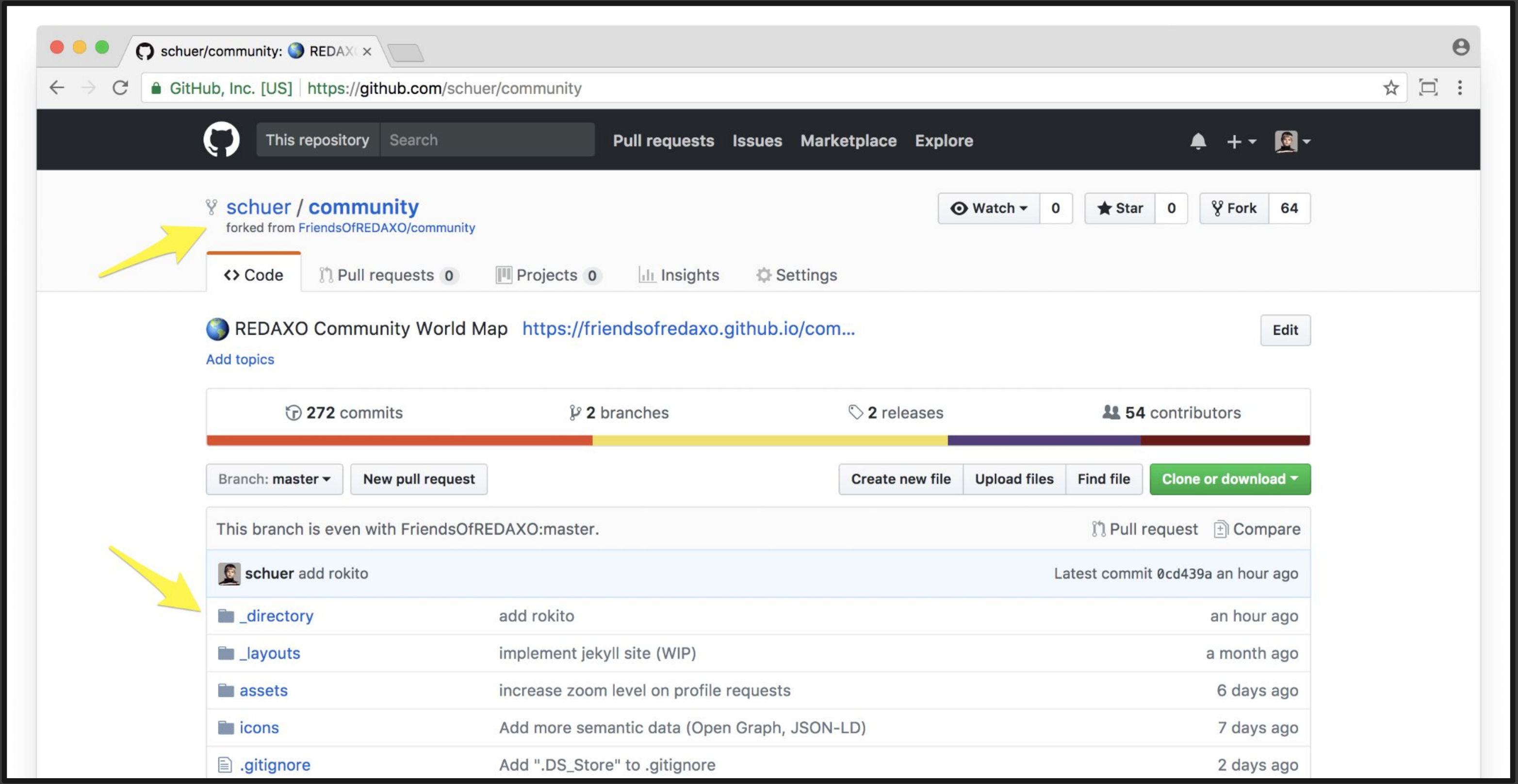
GETTING STARTED – TOOLS

CHROME BROWSER

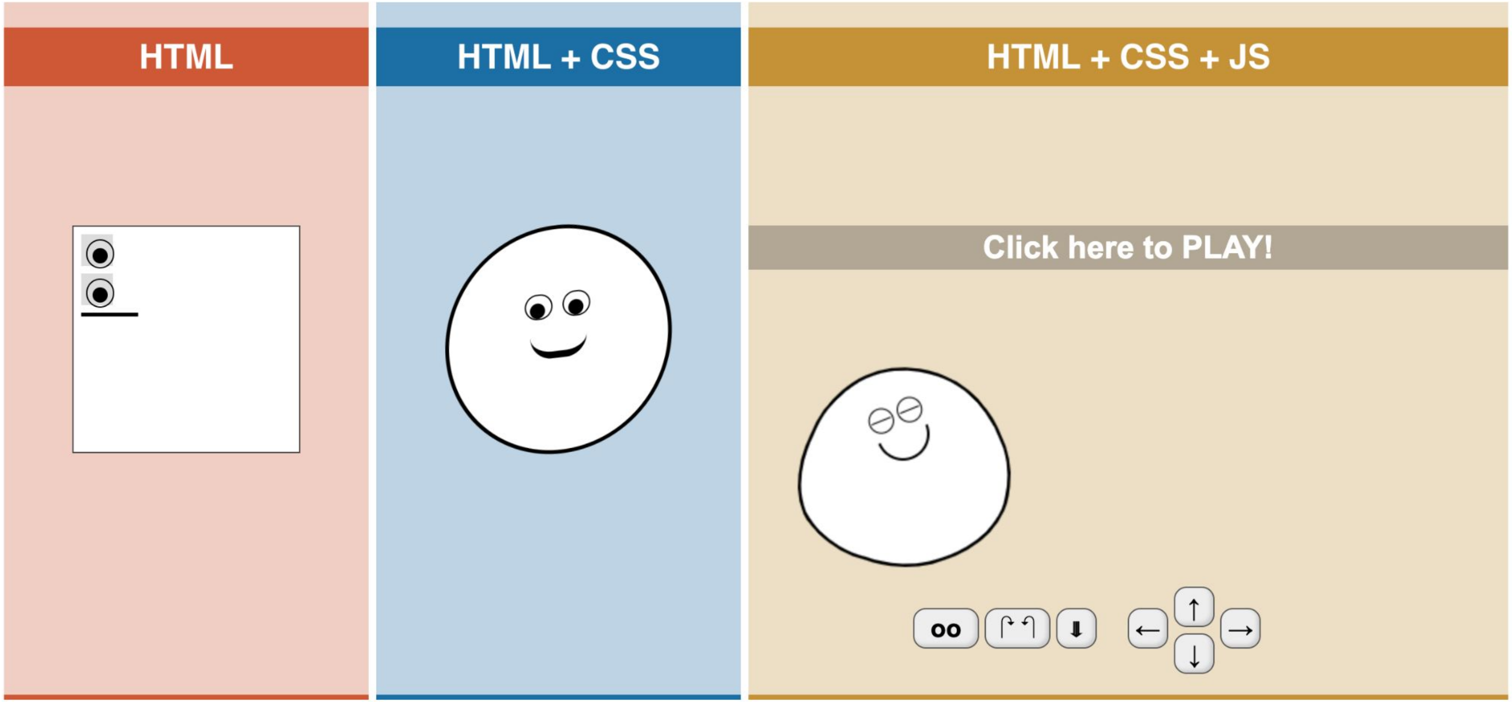


GETTING STARTED – TOOLS

GITHUB ACCOUNT



HOW A WEBSITE IS MADE



WHAT IS HTML?

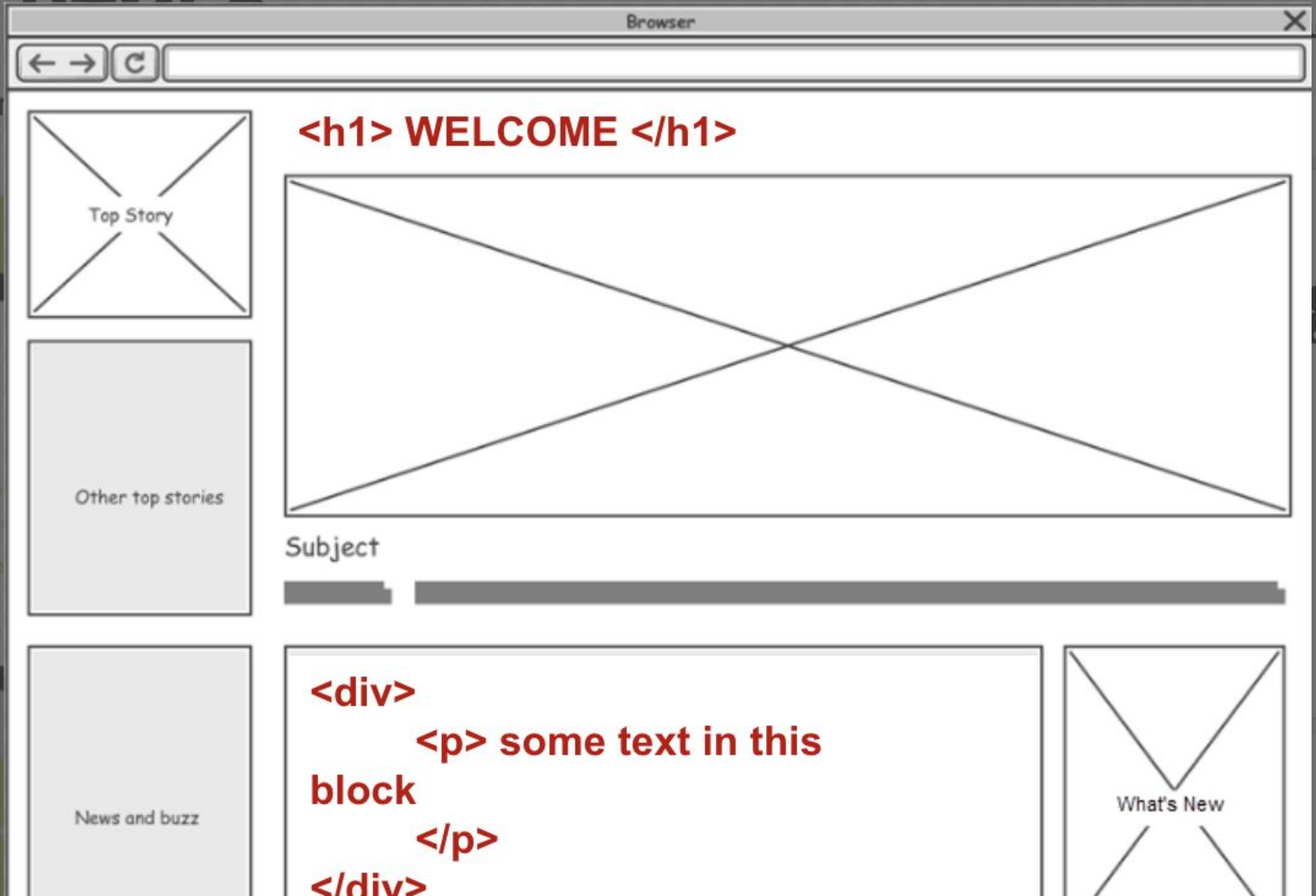
HTML (Hypertext Markup Language) is the standard markup language, whose primary function is to provide **structure** to websites.

```html
<!DOCTYPE html>
<html>
<title>HTML Tutorial</title>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

WHAT IS HTML?

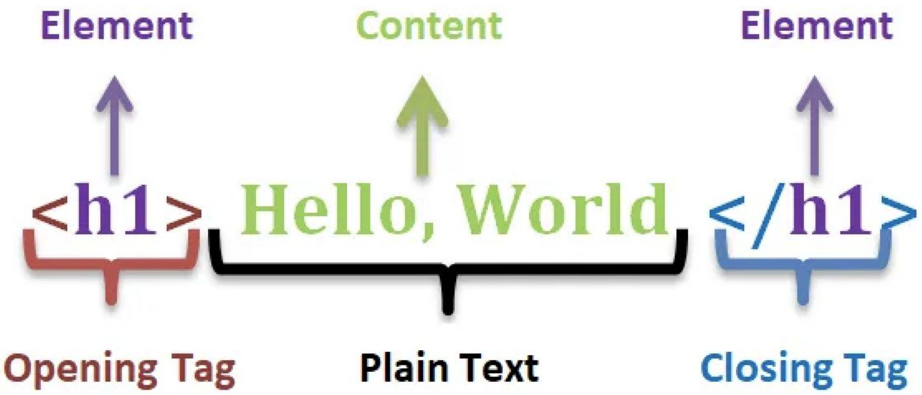
HTML (Hypertext ure to websites.

```
<!DOCTYPE htm
<html>
<title>HTML T
<body>

<h1>This is a
<p>This is a

</body>
</html>
```



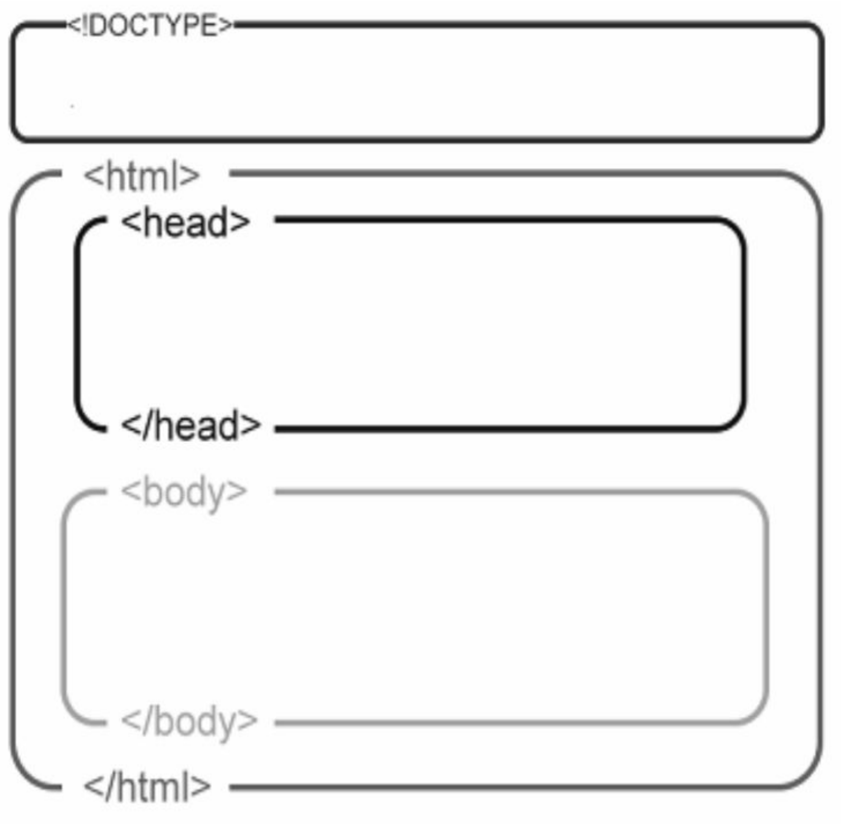
ELEMENTS

The building blocks of HTML



STRUCTURE OF AN HTML PAGE

A HTML page outline is structured in a particular way



<!DOCTYPE html>

Tells the browser what type of code (HTML) to expect and which version (5) we want to use

<html>

All HTML we wish for the browser to pick up must be within the <html> tags

<head>

The "thinking" part of our HTML where we bring in fonts, styles, javascript etc

<body>

Where we have our actual content

LET'S CREATE OUR FIRST WEB PAGE!

USE VISUAL STUDIO APP

Hint: Notes on the steps

MORE ELEMENTS

HEADERS AND CONTENT

Headers

```
<!-- headers 1- 6 -->
<h1>Header 1</h1>
<h2>Header 2</h2>
<h3>Header 3</h3>
<h4>Header 4</h4>
<h5>Header 5</h5>
<h6>Header 6</h6>
```

Header 1

Header 2

Header 3

Header 4

Header 5

Header 6

Paragraphs and span

```
<!-- paragraph and span-->
<p>
  Lorem ipsum, dolor sit amet consectetur adipisicing elit. Nam corporis inventore aperiam voluptates numquam
  debitis esse optio eos, sit exercitationem quos quasi eaque <span>blanditiis fuga incidunt</span> in commodi sunt
  saepe!
</p>
```

LISTS

HTML provides three types of lists:

Unordered HTML List

We use the <ul> tag. Each list item starts with the <li> tag.

```
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

→

- Coffee
- Tea
- Milk

Ordered HTML List

We use the <ol> tag. Each list item starts with the <li> tag.

```
<ol>
<li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

→

1. Coffee
2. Tea
3. Milk

HTML Description Lists

HTML also supports description lists.

A description list is a list of terms, with a description of each term.

The <dl> tag defines the description list, the <dt> tag defines the term (name), and the <dd> tag describes each term:

```
<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
</dl>
```

→

Coffee
    - black hot drink
Milk
    - white cold drink

HTML Attributes

- All HTML elements can have attributes
- Attributes provide additional information about elements
- Attributes are **always** specified in the **start** tag
- Attributes usually come in name/value pairs like: **name="value"**

```
<img src="img_girl.jpg" />
```

The <img> tag is used to embed an image in an HTML page. The src attribute specifies the path to the image to be displayed:

```
<img src="img_girl.jpg" width="500" height="600" />
```

The <img> tag should also contain the width and height attributes, which specify the width and height of the image (in pixels):

```
<p style="color:red;">This is a red paragraph.</p>
```

The style attribute is used to add styles to an element, such as color, font, size, and more.

LINKS

URL

Absolute

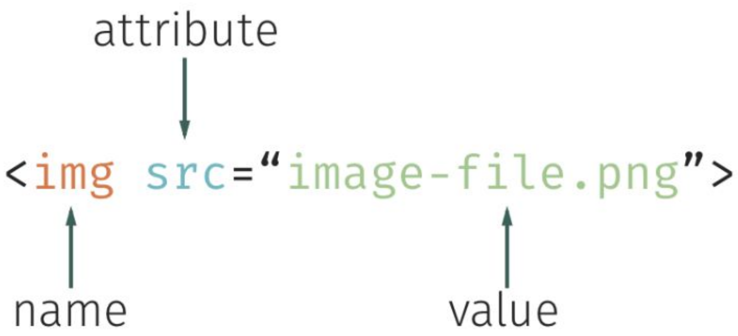
Relative

```html
<!-- links (target="_blank" opens a new tab) -->
<!-- URL path -->
<a href="http://www.google.com" target="_blank">Go to google</a>
<!-- absolute path -->
<a href="/Users/username/Desktop/Code First Girls/Week 1 - HTML/starter-code/pages/page2.html">
  Go to page 2
</a>
<!-- relative path -->
<a href="./pages/page2.html">Go to page 2</a>
```

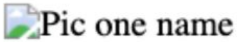
Go to google Go to page 2 Go to page 2

Absolute - typical address format (eg. 123 Main St, London, England, E17 8DN)
Relative - directions you would give someone from A to B (go out of this road, turn left, go down 2 streets etc)
URL - a web address, must start in http

IMAGES




```
<img src="./img/pic1.jpeg" alt="Pic one name">
```



Src is the source of the image - where is it located in the project folder/on the web?
Alt displays text when the image cannot be loaded. It’s also used to make web pages more accessible (for those who use screen readers to know what the image is supposed to be)

HTML Comments

HTML comments are not displayed in the browser, but they can help document your HTML source code.

```
<!-- This is a comment -->

<p>This is a paragraph.</p>

<!-- Remember to add more information here
-->
```

TABLES

Tables are not often used in designing web pages and can be very confusing, so many developers avoid it.

<table> is the tag that allows you create a table.
<thead> is the header row at the top. This makes it bold
<tbody> is for the rest of the columns and rows
<tr> represents a table row
<td> stands for table data and once you have created rows, you can include columns within each row using this tag

```
<table>
  <thead>
    <tr>
      <th>Col 1</th>
      <th>Col 2</th>
      <th>Col 3</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Col 1</td>
      <td>Col 2</td>
      <td>Col 3</td>
    </tr>
    <tr>
      <td>Col 1</td>
      <td>Col 2</td>
      <td>Col 3</td>
    </tr>
    <tr>
      <td>Col 1</td>
      <td>Col 2</td>
      <td>Col 3</td>
    </tr>
  </tbody>
</table>
```

Col 1	**Col 2**	**Col 3**
Col 1	Col 2	Col 3
Col 1	Col 2	Col 3
Col 1	Col 2	Col 3

Name	Age	Class
Steve	40	P.7
Mark	21	S.1

PRACTICE

Exercise

1. Create a new HTML file called **practice.html**
2. Add a main **heading** with your name.
3. Add a **paragraph** about your favorite hobby.
4. Create a **list** of three things you want to learn about web development.
5. Add a **link** to a website you visit often.
6. If you have an **image** file, add it to your page.

HTML FORMS

An **HTML form** is used to collect information from users. The information entered into a form can be sent to a server for processing, such as creating an account, logging in, placing an order, or sending a message.

Some common examples of forms include:

- Login forms
- Registration forms
- Contact forms
- Survey forms
- Job application forms

Basic Form Structure

Every form begins with the `<form>` element.

The <form> Element

The `<form>` tag acts as a container for all form controls.

Example:

```
<form>
    <input type="text">
</form>
```

Common attributes

Attribute	Purpose
`action`	Specifies where the form data is sent.
`method`	Specifies how the data is sent (`GET` or `POST`).

```
<form action="/submit" method="POST">
```

Labels

A **label** describes what information the user should enter.

```
<label for="name">Full Name</label>
<input type="text" id="name">
```

Input Fields

Common Input Types

Text - Used for short text. | <input type="text">

Email - accepts emails | <input type="email">

Password - Hides the characters as the user types. - <input type="password">

SEMANTIC HTML

Semantic HTML uses tags that describe their meaning, not just their appearance. This makes your code more readable and accessible.

Why use Semantic HTML

- Better for SEO(Search Engine Optimization)
- Easier to maintain (clearer code structure)
- Accessibility (screen readers understand the structure)
- Future-proof (easier to style with CSS)

COMMON SEMANTIC TAGS

<header>

Contains introductory content or navigation:

```
<header>
    <h1>My Website</h1>
    <nav>
        <a href="index.html">Home</a>
        <a href="about.html">About</a>
    </nav>
</header>
```

<nav>

Navigation links:

```
<nav>
    <a href="home.html">Home</a>
    <a href="about.html">About</a>
    <a href="contact.html">Contact</a>
</nav>
```

<main>

The main content of the page (use once per page):

```
<main>
    <h1>Welcome</h1>
    <p>Main content goes here...</p>
</main>
```

<section>

A thematic grouping of content:

```
<section>
    <h2>About Me</h2>
    <p>I'm a web developer...</p>
</section>
```

COMMON SEMANTIC TAGS...

<article>
Independent, self-contained content:

```
<article>
    <h2>Blog Post Title</h2>
    <p>Blog post content...</p>
</article>
```

<footer>
Footer information (copyright, links, etc.):

```
<footer>
    <p>&copy; 2025 My Website</p>
</footer>
```

[Download semantic html Sample]

[W3schools - semantic html]

[HTML Assessment]

BYE BYE

Happy CODING